

Mathematical introduction

Branching, quasi-stationarity, and absorption models

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Standalone Pass-2 restructured chapter

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Chapter 1

Mathematical introduction

1.1 Overview

This chapter introduces the stochastic process classes and analytical methods used throughout the thesis. Its organising distinction is between discrete-time and continuous-time Markov chains. In discrete time, the Galton–Watson process shows how independence converts reproduction into functional iteration. In continuous time, exponential holding times and the generator convert event rates into Kolmogorov equations, with birth–death and birth–death–catastrophe processes as the principal count-valued examples. The intended framework also includes time-inhomogeneous chains and coupled ODE–CTMC systems. The available sources do not yet specify the logistic-speciation or coupled-system models, however, so the corresponding subsections mark those gaps explicitly rather than filling them with unsupported mathematics.

The second half of the chapter is organised by method rather than by process. For discrete chains, probability generating functions turn independent sums into products and branching into composition, while first-step analysis isolates one transition and returns a self-consistent equation. Together these methods lead to the binary Galton–Watson survival recursion and the discrete constant $A(p)$. Only the limiting conditional mean requires that constant here; its product, series, bounds and Koenigs-function theory remain the subject of Chapter A, *The constant $A(p)$* .

For continuous chains, the generator supplies forward and backward equations, while generating functions compress an infinite system of state-probability equations into a first-order partial dif-

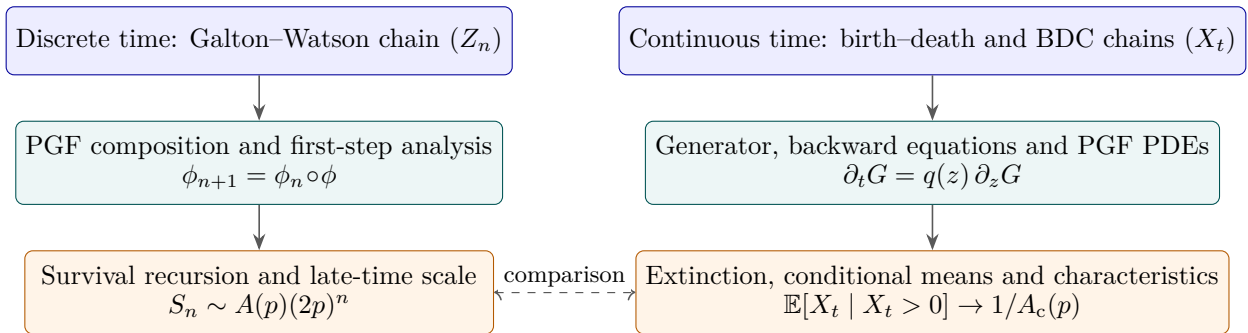


Figure 1.1: The two analytical lanes developed in this chapter. Discrete functional iteration leads to the constant $A(p)$, while continuous-time generator methods give the explicit comparison constant $A_c(p)$ and the PDEs treated by characteristics.

ferential equation. The linear birth–death process serves as the main example because the same model makes the mean, extinction probability, moment calculations and conditioning explicit. In particular, it gives the continuous comparison constant

$$A_c(p) = \frac{1 - 2p}{1 - p},$$

which can be derived in closed form and compared directly with the discrete $A(p)$ without importing the latter’s deeper theory.

The method of characteristics comes last, where it converts a first-order generating-function PDE into ordinary differential equations along characteristic curves. The main text works through one absorption–death model whose solution can also be obtained independently, so each characteristic step can be checked. The broader absorption catalogue is compressed into [appendix A](#), preserving the extension without obscuring the methodological spine of the chapter.

Two further appendices preserve detail that would otherwise overload this introductory route. Appendix [B.1](#) proves the critical Galton–Watson survival and total-progeny asymptotics, while the following parts record the limiting conditional variance, small-founder effects and the discrete rupture calculation. They support the claims used in the main text without transferring the deeper theory of $A(p)$ from Chapter A, *The constant $A(p)$* .

Absorption provides the common thread. Extinction, killing and catastrophe all remove mass from the transient state space, while conditioning asks what remains visible before that absorption occurs. This viewpoint connects elementary first-step arguments to the spectral language of quasi-stationarity and to the generating-function calculations used later in the thesis. Standard background for the continuous-time material is given by Norris [\[1\]](#), Karlin and Taylor [\[2\]](#), and Allen [\[3\]](#); the branching-process background follows Harris [\[4\]](#) and Athreya and Ney [\[5\]](#).

1.2 Markov Chains

A Markov chain is specified by its state space and rules that determine future states from the present one. Its clock may advance in fixed steps or in continuous time, a distinction that changes the analytical machinery without changing the defining principle: conditional on the current state, the past supplies no further information about the future.

1.2.1 Discrete-Time Markov Chains

For a discrete-time chain $(X_n)_{n \geq 0}$ on a countable state space $\mathcal{S}$, the transition probabilities

$$P_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

determine one-step evolution. Repeated matrix multiplication gives multi-step transition probabilities when the state space is conveniently enumerated; for branching processes, generating-function composition is the more natural form of the same iteration.

1.2.1.1 Simple random walks

Simple random walks are intended to provide the elementary discrete-time example and a first setting for hitting-probability problems. The reviewed drafts, notes and bibliography do not contain a substantive treatment, however, so this subsection currently marks the source gap without supplying an unsourced transition law, gambler’s-ruin calculation or citation. An approved treatment can be inserted here without changing the surrounding architecture.

1.2.1.2 Galton–Watson processes

The Galton–Watson process is a discrete-time Markov chain on the non-negative integers. If Z_n is the size of generation n and $L_1, L_2, \dots$ are independent copies of an offspring random variable L , then

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i. \quad (1.1)$$

Writing $\phi(z) = \mathbb{E}(z^L)$ and $\phi_n(z) = \mathbb{E}(z^{Z_n})$, independence gives $\phi_{n+1} = \phi_n \circ \phi$. Thus, if $u_n = \mathbb{P}(Z_n = 0)$ for a process started from $Z_0 = 1$,

$$u_{n+1} = \phi(u_n), \quad u_0 = 0. \quad (1.2)$$

The binary law used throughout the thesis is

$$\mathbb{P}(L = 2) = p, \quad \mathbb{P}(L = 0) = 1 - p, \quad \phi(z) = pz^2 + 1 - p. \quad (1.3)$$

Its mean offspring number is $2p$, so

$$\mathbb{E}(Z_n) = (2p)^n. \quad (1.4)$$

With $S_n = \mathbb{P}(Z_n > 0) = 1 - u_n$, (1.2) becomes

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (1.5)$$

The present framework needs only this definition, its one-step structure and the survival recursion. Two consequences are nevertheless worth recording. At the critical value $p = \frac{1}{2}$,

$$S_n \sim \frac{2}{n}, \quad \mathbb{P}(T = n + 1) \sim \frac{2}{n^2}, \quad (1.6)$$

where $T = \inf\{n : Z_n = 0\}$, while the total progeny $\mathcal{T} = \sum_{n \geq 0} Z_n$ obeys an exact Catalan law and has a $k^{-3/2}$ mass-function tail. The short reciprocal-increment proof and the tree count are collected in [appendix B.1](#). Discrete functional iteration returns briefly in [section 1.3.1](#), before Chapter A, *The constant $A(p)$* develops the fuller Koenigs-function treatment of the constant $A(p)$.

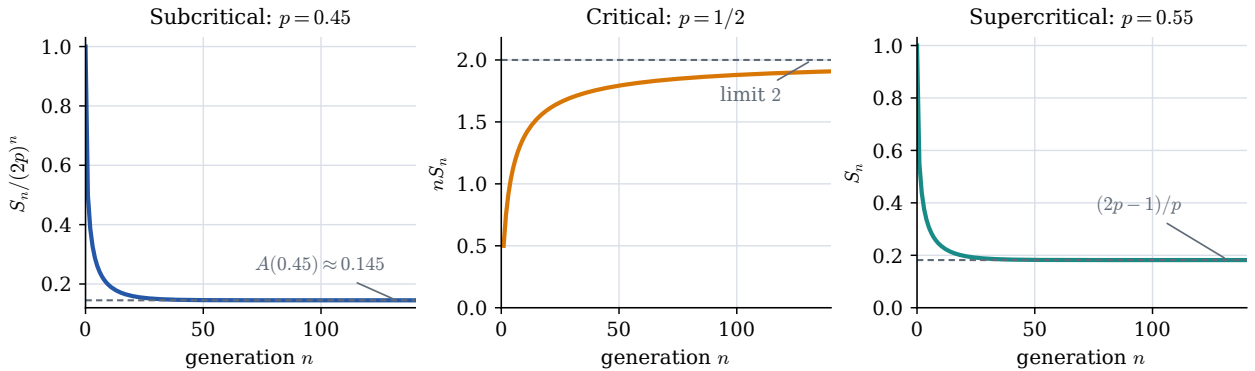


Figure 1.2: Three regime-specific diagnostics for the binary survival recursion (1.5). In the subcritical case, removal of the geometric factor reveals the amplitude $A(p)$; at criticality, nS_n approaches 2; and in the supercritical case, survival approaches the positive fixed point $(2p - 1)/p$. Each curve is obtained by direct iteration from $S_0 = 1$.

1.2.1.3 Founding bottlenecks and conditioned histories

Asymptotic criticality does not remove the early bottleneck faced by a small founding population. At $p = \frac{1}{2}$, the first four extinction probabilities are

$$u_1 = 0.500, \quad u_2 = 0.625, \quad u_3 = 0.6953125, \quad u_4 = 0.7417297 \dots$$

For k independent founders, all k descendant trees are extinct by generation n with probability u_n^k , and hence

$$S_n^{(k)} = 1 - u_n^k = 1 - (1 - S_n)^k. \quad (1.7)$$

This identity separates the one-lineage dynamics from the effect of the founding size. It also makes the relevant selection effect visible: conditioning an ensemble on late survival enriches it for paths with unusually favourable early growth, even though every path was generated by the same offspring law. This “push of the past” is a survivorship effect rather than a change in mechanism [6]. The early-generation calculation, cohort comparison and its interpretation are developed in [appendix B.3](#).

1.2.1.4 A discrete rupture bridge

The same pathwise viewpoint supplies a discrete precursor of catastrophe. If each individual initiates rupture independently with probability κ per generation, and $H_n = \sum_{i=0}^{n-1} \tilde{Z}_i$ is the accumulated exposure of the catastrophe-free genealogy, then

$$\mathbb{P}(\text{no rupture before } n) = \mathbb{E}\left((1 - \kappa)^{H_n}\right) \geq (1 - \kappa)^{\mathbb{E}(H_n)}. \quad (1.8)$$

The inequality is Jensen’s, not a mean-field identity. In particular, it bounds non-rupture and does not by itself bound the joint event of non-rupture and population survival. The deterministic doubling calculation and the exact random-exposure formula appear in [appendix B.4](#).

1.2.2 Continuous-Time Markov Chains

Let $(X_t)_{t \geq 0}$ take values in a countable state space $\mathcal{S}$. A time-homogeneous continuous-time Markov chain is specified by its generator $Q = (q_{ij})$, where

$$q_{ij} = \lim_{h \downarrow 0} \frac{\mathbb{P}(X_{t+h} = j \mid X_t = i)}{h} \quad (j \neq i), \quad q_{ii} = -\sum_{j \neq i} q_{ij} =: -q_i. \quad (1.9)$$

The holding time in state i is exponential with rate q_i , after which the chain jumps to j with probability q_{ij}/q_i . An exponential random variable T of rate θ has

$$\mathbb{P}(T > t) = e^{-\theta t}, \quad \mathbb{E}(T) = \theta^{-1},$$

and is memoryless. If independent clocks of rates $\theta_1, \dots, \theta_m$ compete, their minimum has rate $\Theta = \sum_i \theta_i$ and clock k wins with probability

$$\mathbb{P}\left(T_k = \min_i T_i\right) = \frac{\theta_k}{\Theta}. \quad (1.10)$$

This rate-to-probability conversion supplies the interface between discrete and continuous birth–death descriptions and underlies direct trajectory simulation [7].

If $P(t) = (p_{ij}(t))$ is the transition matrix, the forward Kolmogorov equation is

$$\frac{dP}{dt} = PQ. \quad (1.11)$$

For a fixed initial distribution, its component form is the master equation

$$\frac{dp_j(t)}{dt} = \sum_{k \neq j} q_{kj} p_k(t) - q_j p_j(t). \quad (1.12)$$

The first term collects probability entering j , whereas the second records all probability leaving it.

1.2.2.1 Poisson processes

A Poisson-process introduction is intended to precede the general birth–death examples. The reviewed sources support the exponential holding-time and competing- clock machinery above, but provide no substantive construction, counting law or citation for the associated counting process. This subsection therefore records that source boundary and awaits an approved account; no further axioms or formulae are introduced here.

1.2.2.2 Birth–death processes

In the linear birth–death process, each of the n individuals present contributes independently to birth and death. The non-zero rates from state n are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad (1.13)$$

and 0 is absorbing. The state probabilities satisfy

$$\frac{dp_n}{dt} = \lambda(n-1)p_{n-1} + \mu(n+1)p_{n+1} - (\lambda + \mu)np_n. \quad (1.14)$$

Equivalently, the generator acts on a test function f by

$$(\mathcal{L}f)(n) = \lambda n \{f(n+1) - f(n)\} + \mu n \{f(n-1) - f(n)\}. \quad (1.15)$$

Choosing an indicator function recovers one column of the master equation, while choosing a power or falling factorial produces a moment equation. These uses are developed in [section 1.3.2](#).

The rates also give a direct pathwise construction. Conditional on $X_t = n > 0$, the next event time is exponential with parameter $n(\lambda + \mu)$. Over a short interval h ,

$$\begin{aligned} \mathbb{P}(X_{t+h} = n+1 \mid X_t = n) &= \lambda n h + o(h), \\ \mathbb{P}(X_{t+h} = n-1 \mid X_t = n) &= \mu n h + o(h), \\ \mathbb{P}(X_{t+h} = n \mid X_t = n) &= 1 - n(\lambda + \mu)h + o(h). \end{aligned} \quad (1.16)$$

These relations determine both the jump chain and its clock. They also explain why the model has two complementary descriptions. At population level it is a nearest-neighbour chain with state-dependent speed; at individual level it is the sum of independent descendant families, each begun by one founder.

Let $X_t^{(r)}$ denote the descendants at time t of founder r . Starting from N individuals,

$$X_t = \sum_{r=1}^N X_t^{(r)}, \quad X_t^{(1)}, \dots, X_t^{(N)} \text{ independent and identically distributed.} \quad (1.17)$$

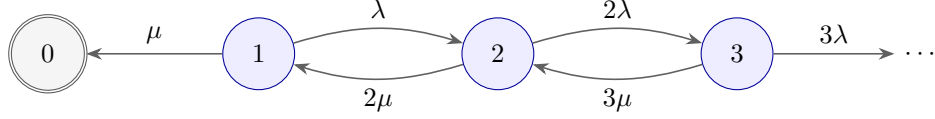


Figure 1.3: The linear birth–death chain. Per-individual rates make both the total event rate and the speed of the clock proportional to the current population; state 0 is absorbing.

This decomposition is stronger than a statement about means. If $F(z, t) = \mathbb{E}(z^{X_t} \mid X_0 = 1)$, then

$$\mathbb{E}(z^{X_t} \mid X_0 = N) = F(z, t)^N. \quad (1.18)$$

Thus the law for every fixed initial population is determined by the one-founder law. Extinction by time t is the specialisation $z = 0$, so its probability from N founders is the N th power of the one-founder probability. Differentiation at $z = 1$ instead shows that the mean is N times the one-founder mean. These are two consequences of the same factorisation, not separate assumptions.

The total event rate in state n is $n(\lambda + \mu)$, so the holding time has mean $\{n(\lambda + \mu)\}^{-1}$. Conditional on a jump, its direction is independent of n :

$$\mathbb{P}(n \rightarrow n + 1 \mid \text{a jump}) = \frac{\lambda}{\lambda + \mu}, \quad \mathbb{P}(n \rightarrow n - 1 \mid \text{a jump}) = \frac{\mu}{\lambda + \mu}. \quad (1.19)$$

Thus the embedded jump chain records the relative rates, while the continuous clock accelerates as the population grows. The independence of descendant families also gives a branching property: if $F(z, t)$ is the PGF from one founder, then the PGF from N founders is $F(z, t)^N$.

The embedded chain alone cannot answer time-dependent questions. For example, two processes with rates (λ, μ) and $(c\lambda, c\mu)$ have the same jump directions but different holding times: multiplying both rates by c accelerates the continuous trajectory by the same factor. Conversely, changing the ratio λ/μ changes the drift of the event sequence itself. This separation is useful in simulation and analysis. One first samples a wait of rate $n(\lambda + \mu)$ and then samples the jump direction using (1.19); analytically, first-step equations perform exactly the same two-stage conditioning.

For the probability generating function $G(z, t) = \sum_{n \geq 0} p_n(t) z^n$, index shifts turn this infinite system into

$$\frac{\partial G}{\partial t} = (\lambda z - \mu)(z - 1) \frac{\partial G}{\partial z}, \quad G(z, 0) = z^N. \quad (1.20)$$

To see where the polynomial coefficient comes from, multiply (1.14) by z^n and sum over $n \geq 0$. The birth inflow, death inflow and total outflow become, respectively,

$$\lambda z^2 G_z, \quad \mu G_z, \quad -(\lambda + \mu) z G_z. \quad (1.21)$$

Their sum factorises as $(\lambda z - \mu)(z - 1)G_z$. The root $z = 1$ expresses conservation of total probability, since (1.20) then gives $\partial_t G(1, t) = 0$. The other root, $z = \mu/\lambda$, encodes the extinction threshold and reappears in the characteristic solution.

The PGF retains the full state law. At any fixed time,

$$p_n(t) = \frac{1}{n!} \left. \frac{\partial^n G}{\partial z^n} \right|_{z=0}, \quad p_0(t) = G(0, t), \quad \mathbb{E}(X_t) = G_z(1, t). \quad (1.22)$$

The points $z = 0$ and $z = 1$ therefore answer different questions: the first extracts absorption at zero, while derivatives at the second extract moments of the unabsorbed distribution. The characteristic solution in section 1.3.2.6 makes both evaluations explicit. This process becomes the main example in section 1.3.2, where the same equation yields moments, extinction probabilities and conditional means.

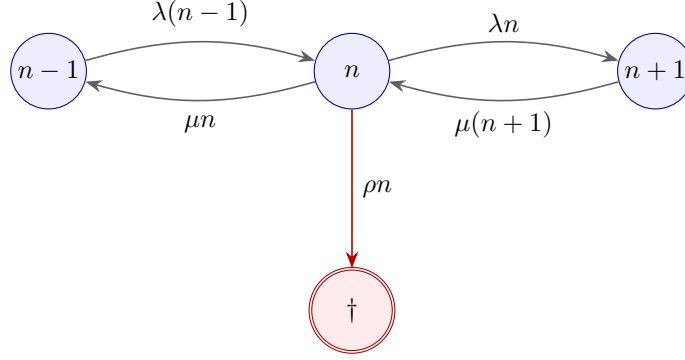


Figure 1.4: Local transition structure of the linear birth–death–catastrophe chain. The horizontal birth and death moves repeat on the non-negative integers; 0 and the catastrophe state $\dagger$ are absorbing.

1.2.2.3 Birth–death–catastrophe processes

A birth–death–catastrophe chain adds an abrupt transition to a distinct absorbing state $\dagger$. For the linear version used later, the rates from $n \geq 1$ are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad q_{n,\dagger} = \rho n, \quad (1.23)$$

with both 0 and $\dagger$ absorbing. The two absorbing mechanisms are not interchangeable. Ordinary extinction reaches 0 through successive deaths, whereas catastrophe removes the current population in one event. On the non-catastrophe states, the sub-probability generating function

$$G(z, t) = \mathbb{E}\left(z^{X_t} \mathbf{1}\{X_t \neq \dagger\}\right)$$

satisfies

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda + \mu + \rho)z + \mu) \frac{\partial G}{\partial z}, \quad G(z, 0) = z^N. \quad (1.24)$$

Unlike (1.20), this PDE need not conserve $G(1, t)$, because its missing mass is precisely the probability already removed by catastrophe. A population-dependent catastrophe rate also reweights whole paths through their accumulated exposure and therefore cannot be replaced by a hazard evaluated at the mean population.

Restricting the generator to the positive states makes the absorption mechanism especially clear. For general birth, death and catastrophe rates $\lambda_i, \mu_i, \kappa_i$, the killed subgenerator K has

$$K_{i,i+1} = \lambda_i, \quad K_{i,i-1} = \mu_i \ (i \geq 2), \quad K_{i,i} = -(\lambda_i + \mu_i + \kappa_i). \quad (1.25)$$

The missing row mass is the instantaneous absorption rate. If T is the time of either ordinary extinction or catastrophe, summing the positive-state forward equations gives

$$\frac{d}{dt} \mathbb{P}(T > t) = -\mu_1 \mathbb{P}(X_t = 1) - \sum_{i \geq 1} \kappa_i \mathbb{P}(X_t = i), \quad (1.26)$$

where the probabilities refer to the chain before absorption. Ordinary death removes survival mass only from state 1; catastrophe may remove it from every positive state.

If ν is quasi-stationary for the killed chain, then for some $\theta > 0$

$$\nu K = -\theta \nu, \quad \mathbb{P}_\nu(T > t) = e^{-\theta t}, \quad \theta = \mu_1 \nu_1 + \sum_{i \geq 1} \kappa_i \nu_i. \quad (1.27)$$

A constant catastrophe clock merely shifts the survival decay rate and leaves the law conditional on survival unchanged. State-dependent catastrophe instead reweights paths by their accumulated exposure, precisely the distinction already visible in (1.8).

1.2.3 Time-Inhomogeneous Processes

Time-inhomogeneous chains allow their transition rules to depend explicitly on calendar time. The intended thesis example is a logistic speciation model, but the reviewed sources contain no equations, parameter definitions or citations from which that model can be reconstructed. The subsection therefore remains an explicit source-gap marker until author-approved material supplies its state space, time-dependent rates and logistic mechanism.

1.2.4 Coupled ODE–CTMC Systems

The final process class couples deterministic continuous variables to a continuous-time Markov chain. The ODE dynamics may depend on the chain state, the chain’s transition rates may depend on the ODE variables, or the dependence may operate in both directions. The reviewed sources contain no concrete coupled model, governing ODE, rate functions or citation, so this subsection records only the intended interface. Its mathematical development awaits author-approved source material rather than an invented schematic example.

1.3 Methods for Markov Chains

The same stochastic process can be analysed at several levels. Transition matrices and generators describe the full law, first-step equations isolate a particular hitting event, and moments compress the law into summary quantities. Generating functions occupy the useful middle ground: they retain enough of the distribution to turn discrete branching into iteration and continuous-time master equations into PDEs. The methods are ordered below by the clock on which the process evolves.

1.3.1 Discrete-Time Methods

For a non-negative integer-valued random variable X , its probability generating function is

$$\phi(z) = \mathbb{E}(z^X) = \sum_{k \geq 0} \mathbb{P}(X = k) z^k. \quad (1.28)$$

Its coefficients determine the distribution, while derivatives at $z = 1$ give factorial moments. For branching, the more important property is structural: independent sums become products and a random sum becomes a composition. The Galton–Watson recursion (1.1) therefore reduces to the one-dimensional iteration (1.2).

First-step analysis provides the corresponding local method. Conditioning on the first transition makes the same unknown quantity reappear from the new state. In the binary process, the founder either leaves no offspring or begins two independent lineages, giving (1.5) directly. The map itself is elementary to iterate, although its late-time matching constant requires a more specialised analysis.

For $p < \frac{1}{2}$, the survival probabilities tend to zero and obey

$$S_n \sim A(p)(2p)^n, \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}. \quad (1.29)$$

Since $\mathbb{E}(Z_n) = (2p)^n$ and $Z_n = 0$ on extinction,

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{(2p)^n}{S_n} \longrightarrow \frac{1}{A(p)}. \quad (1.30)$$

Equations (1.29) and (1.30) provide the complete discrete- $A(p)$ interface needed here. Chapter A, *The constant $A(p)$* develops the product and series representations, bounds, near-critical asymptotics and Koenigs-function methods; that deeper theory is not repeated in this chapter.

The same conditioning has a finite limiting variance. A second-moment recursion for the binary process gives

$$\text{Var}(Z_n \mid Z_n > 0) \longrightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1 - 2p} \right) - \frac{1}{A(p)^2}. \quad (1.31)$$

This is still only a light use of $A(p)$: the moment calculation treats it as the normalising constant already defined in (1.29). The full derivation is given in [appendix B.2](#); its product, series and analytic structure remain exclusively in Chapter A, *The constant $A(p)$* .

1.3.2 Continuous-Time Methods

The generator of a continuous-time chain supports complementary forward and backward viewpoints. The forward equation propagates a distribution, making it the natural starting point for master equations and generating functions. The backward equation instead fixes a terminal state or observable and asks how its value depends on the initial state, which is particularly efficient for hitting and extinction probabilities and for transforms of hitting times.

For a time-homogeneous chain, the transition matrices form a semigroup:

$$P(s + t) = P(s)P(t), \quad P(0) = I. \quad (1.32)$$

The generator is the infinitesimal derivative at zero,

$$Q = \lim_{h \downarrow 0} \frac{P(h) - I}{h}, \quad P(h) = I + hQ + o(h). \quad (1.33)$$

Inserting the short-time expansion into the semigroup relation in its two equivalent orders explains the two Kolmogorov equations. From $P(t + h) = P(t)P(h)$, the short interval acts at the end of the path and gives the forward equation. From $P(t + h) = P(h)P(t)$, it acts at the beginning and gives the backward equation. The matrices coincide as derivatives of the same semigroup, but their componentwise interpretations differ because one resolves the final state and the other resolves the first jump from the initial state.

This distinction prevents a common notational error. With row distributions, $p(t) = p(0)P(t)$ and hence $p'(t) = p(t)Q$. A function of the initial state is naturally a column vector, on which the generator acts from the left. The same entries q_{ij} appear in both calculations, but the object being evolved fixes the multiplication order. All subsequent equations use this convention.

1.3.2.1 Backward equations

In matrix notation the two Kolmogorov equations are

$$\frac{dP(t)}{dt} = P(t)Q, \quad \frac{dP(t)}{dt} = QP(t), \quad (1.34)$$

with the order of multiplication distinguishing forward propagation from backward dependence on the initial state. For a branching process the backward equation can close nonlinearly because independent descendants contribute copies of the same one-ancestor quantity. For example, the PGF $F(x, t)$ of an interior linear birth–death process started from one individual satisfies

$$\frac{\partial F}{\partial t} = \lambda F^2 - (\lambda + \mu)F + \mu, \quad F(x, 0) = x. \quad (1.35)$$

Indeed, during a short initial interval of length h , the founder dies, gives birth, or experiences no event with probabilities μh , λh and $1 - (\lambda + \mu)h$, up to $o(h)$. After a birth, two independent descendant families remain for the residual time. The branching property therefore gives

$$F(x, t + h) = \mu h + \lambda h F(x, t)^2 + \{1 - (\lambda + \mu)h\}F(x, t) + o(h), \quad (1.36)$$

and division by h recovers (1.35). The calculation exposes why the forward equation is linear in the state probabilities while the backward one-founder equation is nonlinear in the PGF.

The generator, master equation and first-step calculation are therefore three representations of the same local transition rules.

1.3.2.2 Generator identities and observables

The generator also evolves observables without first reconstructing the entire distribution. For a suitable function f and a non-explosive chain,

$$\frac{d}{dt} \mathbb{E}(f(X_t)) = \mathbb{E}((\mathcal{L}f)(X_t)). \quad (1.37)$$

This is the expectation form of the backward equation. It follows by applying the infinitesimal transition rule over an interval of length h , conditioning on X_t , and dividing the expected increment by h . For the linear birth–death generator (1.15), the choices $f(n) = n$ and $f(n) = n(n - 1)$ give

$$(\mathcal{L}f)(n) = (\lambda - \mu)n, \quad f(n) = n, \quad (1.38)$$

$$(\mathcal{L}f)(n) = 2(\lambda - \mu)n(n - 1) + 2\lambda n, \quad f(n) = n(n - 1). \quad (1.39)$$

Thus the same moment equations can be obtained either from generator algebra or by differentiating the generating-function PDE. The former is usually quicker for a small number of moments; the latter keeps the whole hierarchy and remains available when a distributional answer is required.

1.3.2.3 Mean population size

For the linear birth–death chain, each individual contributes births at rate λ and deaths at rate μ . Taking the first moment of the master equation, or differentiating (1.20) once at $z = 1$, gives

$$\frac{d}{dt} \mathbb{E}(X_t) = (\lambda - \mu) \mathbb{E}(X_t), \quad \mathbb{E}(X_t) = N e^{(\lambda - \mu)t}. \quad (1.40)$$

The sign of $\lambda - \mu$ separates subcritical, critical and supercritical mean behaviour. This division is useful but incomplete, because the mean alone does not determine extinction or the shape of the state distribution.

1.3.2.4 Variance and higher moments

Generating functions organise the moment hierarchy. At any fixed t ,

$$\left. \frac{\partial G}{\partial z} \right|_{z=1} = \mathbb{E}(X_t), \quad \left. \frac{\partial^2 G}{\partial z^2} \right|_{z=1} = \mathbb{E}(X_t(X_t - 1)), \quad (1.41)$$

so the first two derivatives determine the variance and repeated differentiation gives higher factorial moments. Differentiating the PDE (1.20) in this way replaces the infinite probability system by ordinary differential equations for the moments. If

$$M_1(t) = G_z|_{z=1}, \quad M_2(t) = G_{zz}|_{z=1},$$

then differentiating once and twice before setting $z = 1$ shows explicitly that the first moment closes and that the second factorial moment is driven only by the first two:

$$M_1'(t) = (\lambda - \mu)M_1(t), \quad (1.42)$$

$$M_2'(t) = 2(\lambda - \mu)M_2(t) + 2\lambda M_1(t). \quad (1.43)$$

The variance is $M_2 + M_1 - M_1^2$. Further differentiation produces the same triangular pattern: the equation for the r th factorial moment involves moments of order at most r . This hierarchy is often preferable to solving for every $p_n(t)$ when only a few population summaries are required.

The initial conditions are inherited from the starting population. If $X_0 = N$ deterministically, then $M_1(0) = N$ and $M_2(0) = N(N - 1)$. Consequently (1.42) and (1.43) form a closed, sequential calculation: solve for M_1 , insert that result into the linear equation for M_2 , and only then convert factorial moments to the ordinary variance. This order avoids expanding the master equation term by term and makes clear which parts of the calculation depend on the linear per-capita assumption. Nonlinear rates generally introduce expectations of higher powers and destroy this simple closure.

1.3.2.5 Hitting probabilities

When the target is the probability of reaching one set before another, the full time-dependent law is usually unnecessary. Conditioning on the first exponential clock to ring gives a linear system for the desired probabilities over the transient states. Equivalently, the unknown function is harmonic for the killed generator in the interior, with boundary values fixed on the target sets. This viewpoint covers ordinary extinction at 0 and, for (1.23), the competing event of reaching $\dagger$.

More precisely, let A and B be disjoint absorbing target sets and put $h_i = \mathbb{P}_i(\tau_A < \tau_B)$. Conditioning on the first jump from a transient state i yields

$$h_i = \sum_{j \neq i} \frac{q_{ij}}{q_i} h_j, \quad \text{or equivalently} \quad (Qh)_i = 0, \quad (1.44)$$

with boundary data $h = 1$ on A and $h = 0$ on B . Expected hitting times obey the inhomogeneous companion equation

$$(Qv)_i = -1 \quad (1.45)$$

on the transient states, with $v = 0$ on the target. These equations separate the eventual destination from the holding times: jump probabilities determine (1.44), while the rates also enter (1.45) through the mean wait $1/q_i$.

The same construction handles transforms of hitting times. If $w_i(s) = \mathbb{E}_i[\exp(-s\tau_A); \tau_A < \tau_B]$, the first holding time contributes an additional discount and the interior equation becomes

$$(Qw)_i = sw_i, \quad (1.46)$$

with $w = 1$ on A and $w = 0$ on B . Differentiating at $s = 0$, when justified, returns time moments. Probability, mean time and transform are therefore not separate tricks: they are the homogeneous, Poisson and resolvent versions of the same killed-generator boundary problem.

1.3.2.6 Extinction probabilities

For the linear birth–death chain started from N individuals, the characteristic solution of (1.20) is [8]

$$G(z, t) = \left[\frac{\mu(1-z) - (\mu - \lambda z)e^{-(\lambda-\mu)t}}{\lambda(1-z) - (\mu - \lambda z)e^{-(\lambda-\mu)t}} \right]^N, \quad \lambda \neq \mu. \quad (1.47)$$

The same extinction law follows directly from the backward equation. Setting $q(t) = F(0, t)$ in (1.35) gives

$$q'(t) = \lambda q(t)^2 - (\lambda + \mu)q(t) + \mu, \quad q(0) = 0. \quad (1.48)$$

For $\lambda \neq \mu$ the right-hand side factorises as $\lambda(q-1)(q-\mu/\lambda)$ and separates; at criticality it becomes $\lambda(q-1)^2$. Independence of the N founder families then raises the one-founder extinction probability to the N th power.

The forward PGF gives the same answer through characteristics. Writing $b(z) = (\lambda z - \mu)(z - 1)$, (1.20) is $G_t - b(z)G_z = 0$. Along a characteristic,

$$\frac{dz}{dt} = -b(z), \quad \frac{dG}{dt} = 0. \quad (1.49)$$

For $\lambda \neq \mu$, separation gives the invariant relation

$$\frac{z(t) - 1}{\lambda z(t) - \mu} = e^{(\mu-\lambda)t} \frac{z(0) - 1}{\lambda z(0) - \mu}. \quad (1.50)$$

The initial condition is $G(z, 0) = z^N$, so one solves (1.50) for the foot $z(0)$ of the characteristic and raises that value to the N th power. This inversion produces (1.47). At $\lambda = \mu$ the two roots coalesce; integrating the resulting repeated-root equation gives the second line of (1.51) directly.

Setting $z = 0$ extracts the probability of extinction by time t :

$$p_0(t) = \begin{cases} \left[\frac{\mu - \mu e^{(\mu-\lambda)t}}{\lambda - \mu e^{(\mu-\lambda)t}} \right]^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu. \end{cases} \quad (1.51)$$

Letting $t \rightarrow \infty$ shows that extinction is certain when $\lambda \leq \mu$ and has probability $(\mu/\lambda)^N$ when $\lambda > \mu$.

The three regimes differ even when ultimate extinction is certain. In the subcritical case the survival probability has an exponential tail. At criticality, (1.51) gives

$$\mathbb{P}(X_t > 0 \mid X_0 = 1) = \frac{1}{1 + \lambda t}, \quad (1.52)$$

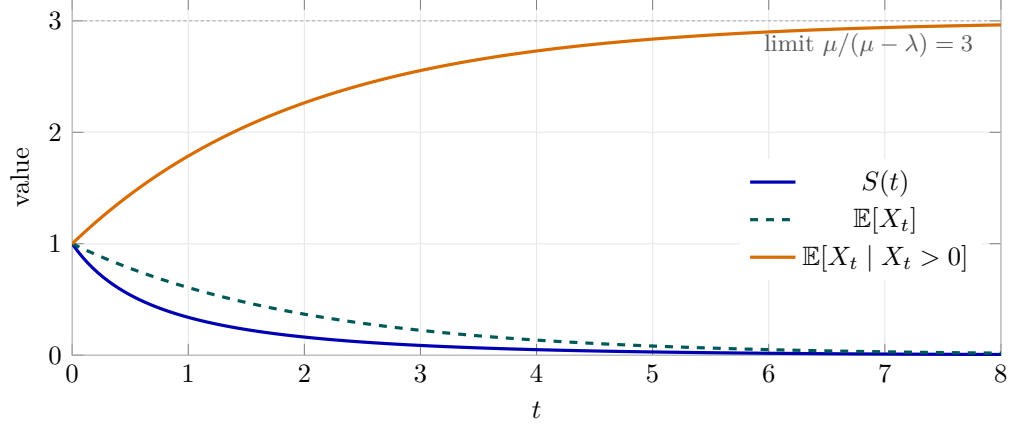


Figure 1.5: Survival probability, unconditional mean and conditional mean for the subcritical linear birth–death process with $\lambda = 1$, $\mu = 3/2$ and $X_0 = 1$. Although both survival and the unconditional mean decay, the mean among surviving realisations approaches a non-zero limit.

the continuous-time counterpart of the n^{-1} decay in (1.6). In the supercritical case a positive fraction of paths escape extinction altogether.

These results also illustrate why an unconditional mean can be misleading near absorption. In the subcritical regime almost all paths are eventually at zero, yet the few paths still alive may contain several individuals. At criticality the mean remains constant while the survival probability decays as t^{-1} , so the surviving population must grow on the reciprocal scale. Conditional means make that balancing mechanism explicit.

1.3.2.7 Conditional means

Conditioning on non-extinction separates the transient population from the mass already absorbed at zero. For one initial individual in the subcritical case $\mu > \lambda$, (1.51) gives

$$S(t) = \mathbb{P}(X_t > 0) = \frac{\lambda - \mu}{\lambda - \mu e^{(\mu - \lambda)t}}, \quad (1.53)$$

whereas (1.40) gives $\mathbb{E}(X_t) = e^{(\lambda - \mu)t}$. Since $X_t = 0$ on extinction,

$$\mathbb{E}(X_t | X_t > 0) = \frac{\mathbb{E}(X_t)}{S(t)} = \frac{\lambda e^{(\lambda - \mu)t} - \mu}{\lambda - \mu} \longrightarrow \frac{\mu}{\mu - \lambda}. \quad (1.54)$$

The comparison with the discrete binary process uses the competing-clock identity (1.10). If the birth and death clocks have rates λ and μ , then the probability that birth occurs first is

$$p = \frac{\lambda}{\lambda + \mu}, \quad \frac{\lambda}{\mu} = \frac{p}{1 - p}. \quad (1.55)$$

Substitution into (1.54) gives

$$\lim_{t \rightarrow \infty} \mathbb{E}(X_t | X_t > 0) = \frac{1 - p}{1 - 2p} = \frac{1}{A_c(p)}, \quad \boxed{A_c(p) = \frac{1 - 2p}{1 - p}}. \quad (1.56)$$

The continuous-time conditional-mean constant is therefore elementary. Its discrete counterpart is the lightly defined $A(p)$ of (1.29), whose reciprocal is the conditional mean. The constants are

not identical; their exact comparison and the obstruction arising from discrete functional iteration are developed in Chapter A, *The constant $A(p)$* .

For completeness, at criticality the same division by (1.52) gives

$$\mathbb{E}(X_t \mid X_t > 0) = 1 + \lambda t, \quad (1.57)$$

so there is no stationary conditional mean at the threshold. For fixed $N > 1$ in the subcritical regime, both the mean and the late survival probability acquire the same leading factor N ; their ratio therefore has the same limit $1/A_c(p)$ as the one-founder process.

1.3.2.8 Conditioning and killed semigroups

Conditional means are one projection of a broader construction. Let T be an absorption time and let K be the generator restricted to the transient states. If the initial distribution on those states is the row vector γ , then the unconditional transient law is

$$\gamma e^{tK}. \quad (1.58)$$

Its total mass is the survival probability $\gamma e^{tK} \mathbf{1}$. Normalising by that mass gives the law seen among realisations that have not yet been absorbed:

$$\mathbb{P}_\gamma(X_t = j \mid T > t) = \frac{(\gamma e^{tK})_j}{\gamma e^{tK} \mathbf{1}}. \quad (1.59)$$

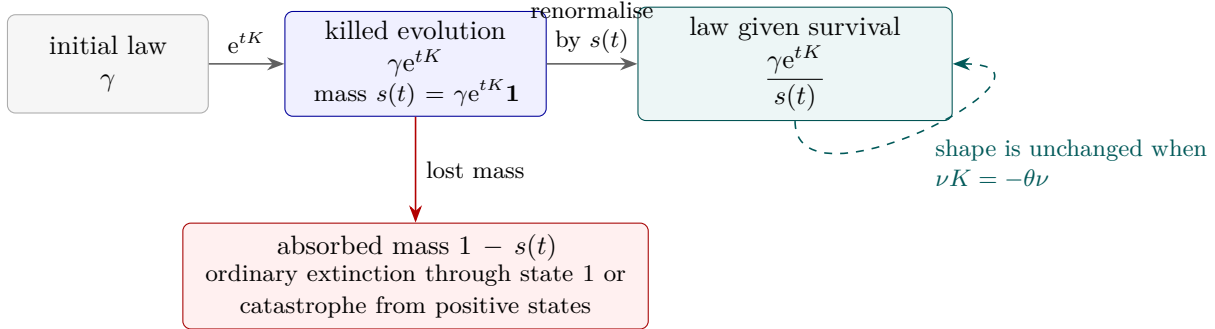


Figure 1.6: Killed evolution and conditioning are distinct operations. The sub-probability law γe^{tK} loses mass to absorption; division by its surviving mass produces the conditional law. A quasi-stationary distribution retains its shape under this renormalised evolution.

Although the unnormalised evolution is linear, this division makes the conditioned evolution nonlinear. The numerator loses mass to absorption while the denominator continually rescales what remains.

A distribution ν is quasi-stationary when the normalised law in (1.59) equals ν for every t if it begins from ν . The eigenrelation

$$\nu K = -\theta \nu \quad (1.60)$$

then gives $\nu e^{tK} = e^{-\theta t} \nu$: survival decays exponentially, but the relative weights of the surviving states do not change. In the BDC process the identity for θ in (1.27) separates the ordinary extinction flux through state 1 from the catastrophe flux across all positive states. State-dependent

killing changes the eigenvector as well as the decay rate; a constant killing clock only shifts the eigenvalue.

Quasi-stationarity is useful even when the process is not started from ν . Under appropriate conditions, the conditional law from a wider class of initial states converges to a Yaglom limit as t grows [9, 10, 11]. The binary Galton–Watson calculation in (1.30) and (1.31) identifies the first two limiting conditional moments without reproving that general convergence theorem. Likewise, the birth–death calculation above obtains the limiting conditional mean directly from an exact survival law.

This viewpoint clarifies the role of the constants. The continuous-time model has an explicit killed evolution, and its limiting conditional mean reduces to $1/A_c(p)$. In discrete time the corresponding normalisation is the late-survival constant $A(p)$. Chapter A, *The constant $A(p)$* studies why functional iteration makes that constant analytically more difficult. The present chapter needs only the probabilistic interface: absorption removes mass, conditioning renormalises the surviving law, and the reciprocal survival amplitude controls its mean.

A method-selection summary

The methods answer different questions from the same generator. Their division of labour for the birth–death example is summarised in table 1.1.

Question	Main object	Governing relation
Full state law	forward probabilities or PGF	$p'(t) = p(t)Q$ or (1.20)
Dependence on initial state	backward observable	$P'(t) = QP(t)$; one-founder equation (1.35)
A few summaries	generator applied to observables	(1.37) and the moment hierarchy
Eventual destination	harmonic boundary problem	$Qh = 0$ with absorbing boundary values
Time to a target	Poisson or resolvent problem	$Qv = -1$ or (1.46)
Law before absorption	killed semigroup and normalisation	(1.59)

Table 1.1: Continuous-time methods organised by the quantity sought. The generator is common to every row; the choice of observable or boundary condition determines the calculation.

1.3.3 Method of Characteristics

A first-order generating-function PDE has the schematic form

$$a_t \frac{\partial G}{\partial t} + a_x \frac{\partial G}{\partial x} + a_y \frac{\partial G}{\partial y} = bG + c.$$

The method of characteristics chooses curves in (x, y, t) with tangent vector (a_x, a_y, a_t) . Along each curve, the chain rule turns the PDE into an ordinary differential equation for G . The calculation proceeds in three stages: derive the PDE from the master equation, integrate the characteristic ODEs from a parametrised initial surface, and invert that parametrisation to recover $G(x, y, t)$. The following example carries out all three stages in a case whose answer can also be obtained from independence.

The initial surface matters. For an evolution problem it is naturally the plane $t = 0$, parametrised by the remaining generating-function variables. A point (x, y, t) is traced backwards along its

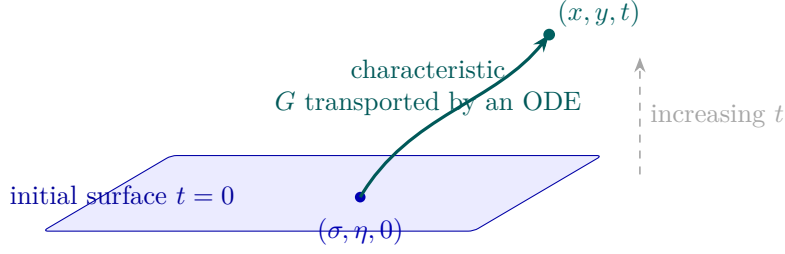


Figure 1.7: Characteristic geometry for an evolution problem. To evaluate the solution at (x, y, t) , trace the characteristic back to its foot $(\sigma, \eta, 0)$ on the initial surface, then apply the initial PGF there.

characteristic until it meets that plane; the initial PGF then supplies the value carried forward along the same curve. The last inversion step is therefore not cosmetic: it identifies which point of the initial surface feeds the requested point at time t .

In practice the calculation is best kept in a fixed order.

1. Derive the PGF PDE from the forward equations, displaying at least one index shift so that signs and boundary terms can be checked.
2. Move all derivative terms to one side and read off the characteristic vector field. A consistent sign convention is essential because reversing the time component reverses every other characteristic equation.
3. Parametrise the initial surface and integrate the characteristic ODEs with those initial values, retaining the labels of the surface coordinates.
4. Invert the map from surface coordinates to (x, y, t) before substituting the initial PGF. This is often the longest algebraic step.
5. Check the initial condition, normalisation at unit PGF arguments, and at least one independent moment or finite-state reduction.

These stages separate probabilistic bookkeeping from differential-equation algebra. They also locate errors: a failure at $G(1, 1, t) = 1$ usually points to the PDE or its signs, whereas a correct PDE but incorrect initial value points to the characteristic inversion.

Worked example: absorption–death

Let X_t and Y_t count particles inside and outside a compartment. Each exterior particle is absorbed at rate α , while each interior particle dies at rate μ . Starting from $X_0 = 0$ and $Y_0 = N$, the transitions give

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \quad (1.61)$$

For

$$G(x, y, t) = \sum_{i,j} p_{i,j}(t) x^i y^j,$$

multiplication by $x^i y^j$, summation and index shifts produce

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (1.62)$$

For example, the absorption inflow term becomes

$$\sum_{i,j \geq 0} \alpha(j+1)p_{i-1,j+1}x^i y^j = \alpha x \frac{\partial G}{\partial y},$$

while absorption outflow contributes $-\alpha y G_y$. The death terms similarly combine to $\mu(1-x)G_x$. This index shift is the bridge from the countable forward system to the first-order PDE; linear per-particle rates are exactly what leave only first derivatives.

Parametrise the initial surface by

$$x(\sigma, \eta, 0) = \sigma, \quad y(\sigma, \eta, 0) = \eta, \quad t(\sigma, \eta, 0) = 0.$$

Writing the PDE with all terms on one side and matching the chain rule gives

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dx}{d\tau} = \mu(1-x), \quad \frac{dy}{d\tau} = \alpha(x-y). \quad (1.63)$$

The first two equations say

$$G = \eta^N, \quad t = -\tau. \quad (1.64)$$

The x -equation gives

$$1-x = (1-\sigma)e^{-\mu\tau}. \quad (1.65)$$

Substitution into the remaining linear equation and use of the initial condition gives

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1-\sigma)e^{-\mu\tau} + \left(\eta + \frac{\alpha}{\alpha - \mu}(1-\sigma) - 1\right)e^{-\alpha\tau}. \quad (1.66)$$

To return from (σ, η, τ) to (x, y, t) , rearrange (1.66), using (1.65) and setting $\tau = -t$ yields

$$\eta = 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y. \quad (1.67)$$

Consequently

$$G(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y\right)^N. \quad (1.68)$$

Independent check and removable equal-rate case

For one exterior particle, only three states are reachable: still outside, absorbed and alive, or dead. Their probabilities are

$$\begin{aligned} p_{0,1}(t) &= e^{-\alpha t}, \\ p_{1,0}(t) &= \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t}), \\ p_{0,0}(t) &= 1 - p_{0,1}(t) - p_{1,0}(t). \end{aligned} \quad (1.69)$$

The corresponding paths through the three one-particle states provide a direct visual check on the PGF coefficients. Thus the one-particle PGF is $p_{0,0} + p_{1,0}x + p_{0,1}y$, and independence raises this expression to the N th power, reproducing (1.68). At the removable degeneracy $\alpha = \mu$, l'Hôpital's rule gives

$$p_{1,0}(t) = \alpha t e^{-\alpha t}, \quad (1.70)$$

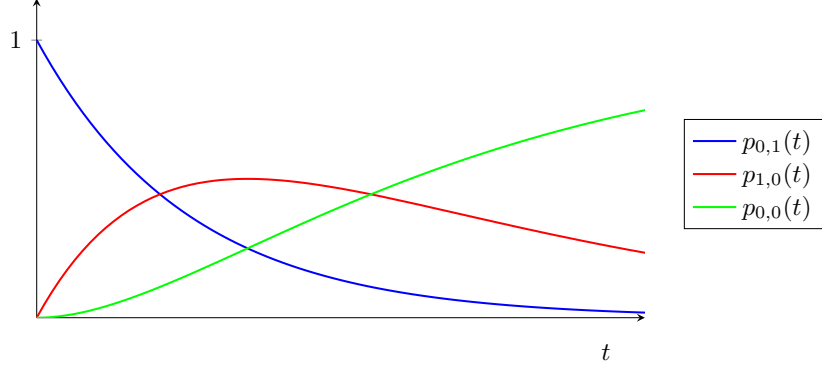


Figure 1.8: State probabilities for the single-particle absorption–death model. The particle begins outside, may be absorbed while alive, and eventually reaches the dead state. These three curves are the coefficients of the one-particle PGF and sum to one at every time.

so the stochastic model remains regular even though the separated expression contains $\mu - \alpha$ in its denominator.

The PGF also recovers the full joint law without another differential equation. Put $c = p_{0,0}(t)$, $f = p_{1,0}(t)$ and $g = p_{0,1}(t)$. Since $G = (c + fx + gy)^N$, coefficient extraction gives

$$\mathbb{P}(X_t = i, Y_t = j) = \frac{N!}{i!j!(N-i-j)!} f^i g^j c^{N-i-j}, \quad i + j \leq N. \quad (1.71)$$

Thus the characteristic solution contains a trinomial distribution over the three possible single-particle states. Differentiation at $(x, y) = (1, 1)$ gives

$$\mathbb{E}(X_t) = Nf(t), \quad \mathbb{E}(Y_t) = Ng(t), \quad \text{Cov}(X_t, Y_t) = -Nf(t)g(t). \quad (1.72)$$

The negative covariance records competition between mutually exclusive states of the same initial particle. These coefficient and moment checks test more than normalisation: they confirm that the characteristic inversion has retained the correct joint dependence.

Several checks now test different parts of the calculation. At $t = 0$ the PGF is y^N ; at $x = y = 1$ it is one; and its coefficients are non-negative state probabilities summing to one. Differentiation verifies the PDE, while the single-particle calculation verifies the characteristic inversion independently. The method is useful precisely because the same steps continue to work when birth opens an infinite interior state space and no finite-state shortcut remains. The absorption-only and absorption–birth–death extensions, together with coefficient recovery, are retained in [appendix A](#).

To see the change caused by birth without adding a second worked derivation, let interior particles reproduce at rate λ as well as die at rate μ . The absorption term is unchanged, but the interior birth transition contributes $\lambda(x^2 - x)G_x$. The PGF equation is therefore

$$G_t = \{\lambda x^2 - (\lambda + \mu)x + \mu\}G_x + \alpha(x - y)G_y. \quad (1.73)$$

The coefficient of G_x is the same factored birth–death polynomial encountered in (1.20). What changes is the state space: reproduction destroys the three-state reduction, so the characteristic calculation is no longer merely an alternative to finite-state independence. Appendix A retains the resulting characteristic system, closed form and coefficient-recovery machinery. The main text thus supplies one complete worked example while keeping the extended model available for later use.

Appendix A

Extended absorption catalogue

The main text uses absorption–death as its single worked characteristics example. This appendix retains the neighbouring models in a more compact form, preserving the progression from a finite independent-particle system to an infinite-state branching system without interrupting the framework-and-methods architecture.

A.1 Absorption only

Let X_t and Y_t count particles inside and outside a compartment. In the absorption-only model, each of the Y_t exterior particles moves inside at rate α . Starting from $(X_0, Y_0) = (0, N)$, the generating function satisfies

$$\frac{\partial G}{\partial t} = \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (\text{A.1})$$

Each initial particle remains outside until an exponential clock of rate α rings, and the particles act independently. Hence

$$G(x, y, t) = \left[(1 - e^{-\alpha t})x + e^{-\alpha t}y \right]^N. \quad (\text{A.2})$$

Thus X_t is binomial with parameters N and $1 - e^{-\alpha t}$, with $Y_t = N - X_t$. In this finite system, the independence argument is simpler than solving the PDE that the same assumptions produce.

A.2 Absorption–birth–death

Now retain absorption at rate α for each exterior particle, and let each interior particle give birth at rate λ and die at rate μ . The generating-function equation becomes

$$\frac{\partial G}{\partial t} = (\lambda x^2 - (\lambda + \mu)x + \mu) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (\text{A.3})$$

An absorbed particle may now initiate an unbounded branching population, so the finite three-state shortcut used in the worked example no longer closes.

A parameter-regular representation follows by conditioning on the absorption time. Let $F(x, t)$ be the one-ancestor interior birth–death PGF solving (1.35). One initial exterior particle either remains outside to time t , or is absorbed and leaves an interior process for the remaining time. Therefore

$$g(x, y, t) = e^{-\alpha t}y + \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv, \quad G(x, y, t) = g(x, y, t)^N. \quad (\text{A.4})$$

This formula remains valid at $\lambda = \mu$ and at resonant parameter values, where a separated special-function expression would require a limiting interpretation. It also makes the normalisation checks immediate:

$$G(x, y, 0) = y^N, \quad G(1, 1, t) = 1, \quad G(1, y, t) = [e^{-\alpha t} y + 1 - e^{-\alpha t}]^N.$$

The fuller derivation expresses the characteristic integral through a Gauss hypergeometric function and treats the associated resonance and analytic-continuation domains. Those details are retained in the donor material but are not needed for the single-method introduction here.

A.3 Recovering state probabilities

For any bivariate PGF analytic at the origin,

$$p_{i,j}(t) = [x^i y^j] G(x, y, t) = \frac{1}{i! j!} \left. \frac{\partial^{i+j} G}{\partial x^i \partial y^j} \right|_{x=y=0}. \quad (\text{A.5})$$

In particular, if

$$G(x, y, t) = (c(t) + f(t)x + g(t)y)^N,$$

then

$$p_{i,j}(t) = \frac{N!}{i! j! (N - i - j)!} f(t)^i g(t)^j c(t)^{N-i-j}, \quad i + j \leq N. \quad (\text{A.6})$$

For absorption–death, the functions c, f, g are the three coefficients inside the parentheses in (1.68), so the joint state law is trinomial. By contrast, (A.4) is not multilinear in x for absorption–birth–death; its remaining univariate coefficients must be recovered by differentiation, truncated-series convolution or Cauchy extraction.

Appendix B

Branching details and small populations

The main text uses the binary Galton–Watson process as a compact discrete-time example. This appendix retains the calculations that support that example but would otherwise interrupt the process-and-methods structure: the critical power laws, the limiting conditional variance, the effect of several founders, and the discrete rupture calculation.

B.1 Critical survival, lifetime and total progeny

At $p = \frac{1}{2}$, the mean population is one at every generation, although extinction is certain. The apparent tension is resolved by a heavy tail: a small fraction of realisations become sufficiently large to hold the unconditional mean at one.

Proposition B.1 (Critical survival probability). *For the critical binary Galton–Watson process,*

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty). \quad (\text{B.1})$$

Proof. At $p = \frac{1}{2}$, the exact recursion (1.5) becomes

$$S_{n+1} = S_n - \frac{1}{2}S_n^2 = S_n \left(1 - \frac{S_n}{2}\right). \quad (\text{B.2})$$

Since $S_n \downarrow 0$, its reciprocal tends to infinity. The reciprocal increments satisfy

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{2(1 - S_n/2)} \rightarrow \frac{1}{2}. \quad (\text{B.3})$$

Stolz–Cesàro applied to $(1/S_n)/n$, equivalently the Cesàro limit of the increments in (B.3), gives $(1/S_n)/n \rightarrow 1/2$. This is (B.1). $\square$

Let

$$T = \inf\{n \geq 0 : Z_n = 0\} \quad (\text{B.4})$$

be the extinction time. Since $T = \sum_{n \geq 0} \mathbf{1}_{\{T > n\}}$, Tonelli’s theorem gives the tail-sum identity

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} S_n. \quad (\text{B.5})$$

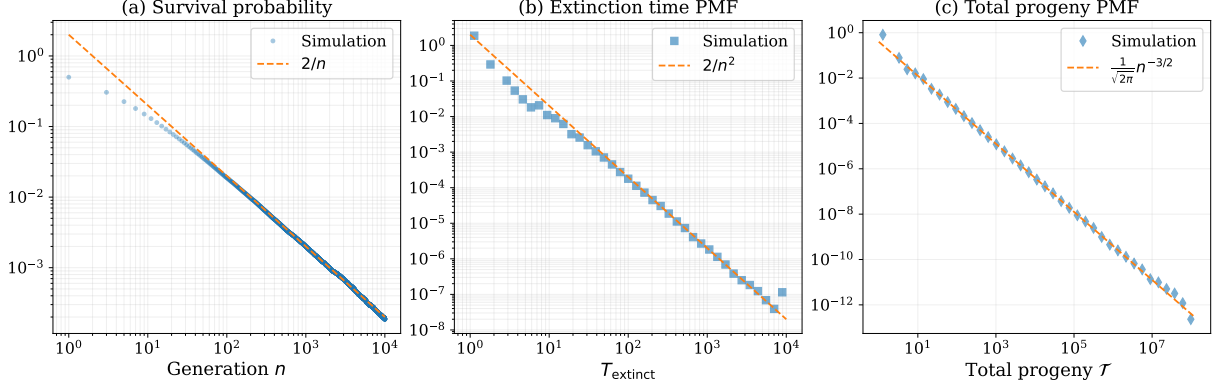


Figure B.1: Critical power laws from 10^6 simulated binary Galton–Watson realisations. From left to right: $S_n \sim 2/n$, $\mathbb{P}(T = n) \sim 2/n^2$, and the $n^{-3/2}$ total-progeny law on its odd support. The first two exponents also follow directly from (B.2).

The last series has the same divergent tail as $2 \sum_n n^{-1}$, and hence $\mathbb{E}(T) = \infty$. Extinction is almost sure, but its expected time is infinite. Moreover, the decrement in (B.2) is exact, so

$$\mathbb{P}(T = n + 1) = S_n - S_{n+1} = \frac{1}{2} S_n^2 \sim \frac{2}{n^2}. \quad (\text{B.6})$$

The total progeny

$$\mathcal{T} = \sum_{n \geq 0} Z_n \quad (\text{B.7})$$

has a different exponent. A finite binary genealogy with k division vertices contains $k + 1$ terminal vertices and hence $2k + 1$ individuals. The number of ordered full binary trees with k division vertices is the Catalan number

$$C_k = \frac{1}{k+1} \binom{2k}{k}.$$

Each such tree has probability $p^k(1-p)^{k+1}$, and therefore

$$\mathbb{P}(\mathcal{T} = 2k + 1) = C_k p^k (1-p)^{k+1}. \quad (\text{B.8})$$

At criticality, $C_k \sim 4^k / (\sqrt{\pi} k^{3/2})$, so Stirling's formula gives

$$\mathbb{P}(\mathcal{T} = 2k + 1) \sim \frac{1}{2\sqrt{\pi}} k^{-3/2}, \quad (\text{B.9})$$

with zero probability at even values. The survival, lifetime-mass and total-progeny exponents are shown together in figure B.1.

B.2 The limiting conditional variance

Let $m = 2p$ and write $M_n = \mathbb{E}(Z_n^2)$. If σ^2 is the offspring variance, then conditional on Z_n the next generation is a sum of Z_n independent offspring counts. Consequently

$$\mathbb{E}(Z_{n+1}^2 \mid Z_n) = \sigma^2 Z_n + m^2 Z_n^2, \quad (\text{B.10})$$

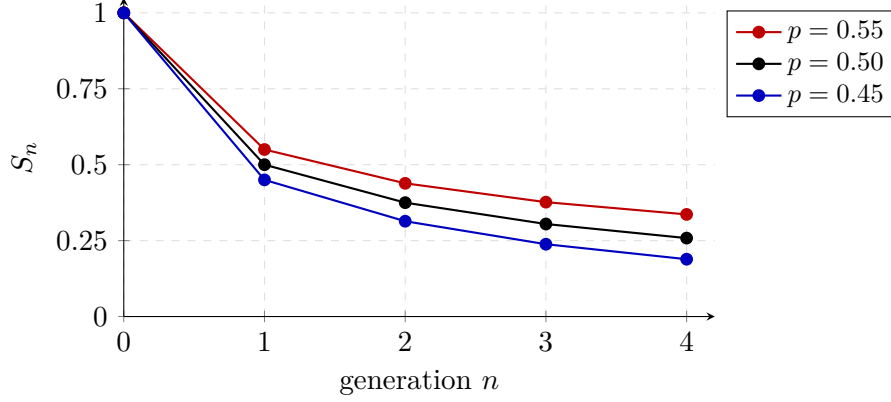


Figure B.2: Survival over the first four generations from one founder. Early loss is steep in all three regimes; a modest supercritical advantage does not remove the founding bottleneck.

and hence

$$M_{n+1} = m^2 M_n + \sigma^2 m^n, \quad M_0 = 1. \quad (\text{B.11})$$

For the binary offspring law, $\sigma^2 = 4p(1-p) = m(2-m)$. Solving (B.11) for $m \neq 1$ gives

$$\begin{aligned} \mathbb{E}(Z_n^2) &= m^{2n} + \sigma^2 m^{n-1} \frac{1-m^n}{1-m} \\ &= m^n \left(1 + \frac{1}{1-m} \right) - \frac{m^{2n}}{1-m}. \end{aligned} \quad (\text{B.12})$$

The second line is specific to the binary variance. At $n = 1$ it returns $\mathbb{E}(Z_1^2) = 4p$, which checks the sign of the final term.

For $m < 1$, divide by $S_n \sim A(p)m^n$ to obtain

$$\mathbb{E}(Z_n^2 \mid Z_n > 0) = \frac{\mathbb{E}(Z_n^2)}{S_n} \longrightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right). \quad (\text{B.13})$$

Combining this with $\mathbb{E}(Z_n \mid Z_n > 0) \rightarrow 1/A(p)$ proves (1.31). These limits are the first two moments of the binary Yaglom law. The moment calculation does not itself prove convergence of the full conditional distribution; that conclusion belongs to the classical quasi-stationary theory developed for Galton–Watson processes [4, 5].

B.3 Small founding populations

For a single founder, the survival and extinction recursions start at $S_0 = 1$ and $u_0 = 0$. At criticality, survival through two generations is

$$S_2 = \frac{1}{2} \left(1 - \frac{1}{4} \right) = \frac{3}{8}. \quad (\text{B.14})$$

The factor $1/2$ requires the founder to divide, while $1 - 1/4$ requires at least one of its two descendants to divide in the next generation. Iteration gives the early extinction probabilities quoted in section 1.2.1.3.

For $Z_0 = k$, the k descendant trees are independent. The exact survival law (1.7) has two useful limits. When S_n is small and $kS_n \ll 1$,

$$S_n^{(k)} = 1 - (1 - S_n)^k \sim kS_n, \quad (\text{B.15})$$

whereas the full expression must be retained when the cohort makes survival appreciable. A larger cohort can postpone extinction at criticality but cannot prevent it; just above criticality it also raises the probability that at least one founding lineage belongs to the non-extinguishing set.

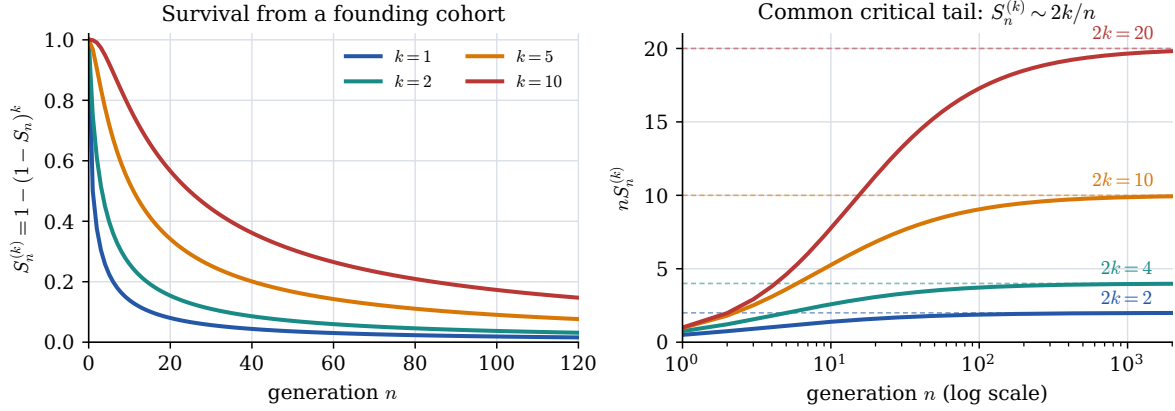


Figure B.3: Critical survival from k independent founders. The left panel shows the exact cohort law (1.7): additional founders soften the early bottleneck but do not prevent eventual extinction. The right panel rescales by generation number; the approach to $2k$ displays the common tail $S_n^{(k)} \sim 2k/n$.

Conditioning on survival to a late time selects paths after they have been generated. Paths with favourable early reproduction are over-represented among the survivors, so their mean early growth can exceed the unconditional mean even when the offspring law is time-homogeneous. This push-of-the-past effect was identified in longevity and diversity data by Budd and Mann [6]. Here it follows directly from the steep early attrition in figures B.2 and B.3: the conditioning changes the sampled ensemble, not the transition mechanism.

B.4 Discrete rupture and the Jensen bound

Suppose first that the population doubles deterministically at each generation until rupture. Each individual exposed during a step independently initiates rupture with probability $\kappa \in [0, 1]$, and a single initiation ends the process. The population exposed during generations $0, \dots, n-1$ has total size

$$1 + 2 + \dots + 2^{n-1} = 2^n - 1.$$

The probability of no rupture before generation n is therefore

$$R_n = (1 - \kappa)^{2^n - 1}, \quad (\text{B.16})$$

and, conditional on reaching step n , the probability of rupture during that step is

$$1 - (1 - \kappa)^{2^{n-1}}. \quad (\text{B.17})$$

When death is allowed, the accumulated exposure is random. Let $\tilde{Z}_i$ denote the binary Galton–Watson population with rupture switched off and define

$$H_n = \sum_{i=0}^{n-1} \tilde{Z}_i. \quad (\text{B.18})$$

Conditioning on the whole catastrophe-free genealogy gives the exact identity

$$\mathbb{P}(\text{no rupture before } n) = \mathbb{E}\left((1 - \kappa)^{H_n}\right). \quad (\text{B.19})$$

The map $h \mapsto (1 - \kappa)^h$ is convex, so Jensen’s inequality yields

$$\mathbb{P}(\text{no rupture before } n) \geq (1 - \kappa)^{\mathbb{E}(H_n)}. \quad (\text{B.20})$$

With $m = 2p$,

$$\mathbb{E}(H_n) = \sum_{i=0}^{n-1} m^i = \begin{cases} \frac{1 - m^n}{1 - m}, & m \neq 1, \\ n, & m = 1. \end{cases} \quad (\text{B.21})$$

Replacing exposure by its mean is thus a lower bound, not an equality.

Finally, non-rupture is not the same event as survival. The probability that the population is both unruptured and non-zero at generation n is

$$\mathbb{E}\left(\mathbf{1}_{\{\tilde{Z}_n > 0\}}(1 - \kappa)^{H_n}\right), \quad (\text{B.22})$$

for which (B.20) does not directly provide a bound. This distinction is the discrete counterpart of separating ordinary extinction from catastrophe in the killed-chain balance (1.26).

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